Zero Sigma Fund Report

Financial Planning and Investment Analysis (1001)

Group 5

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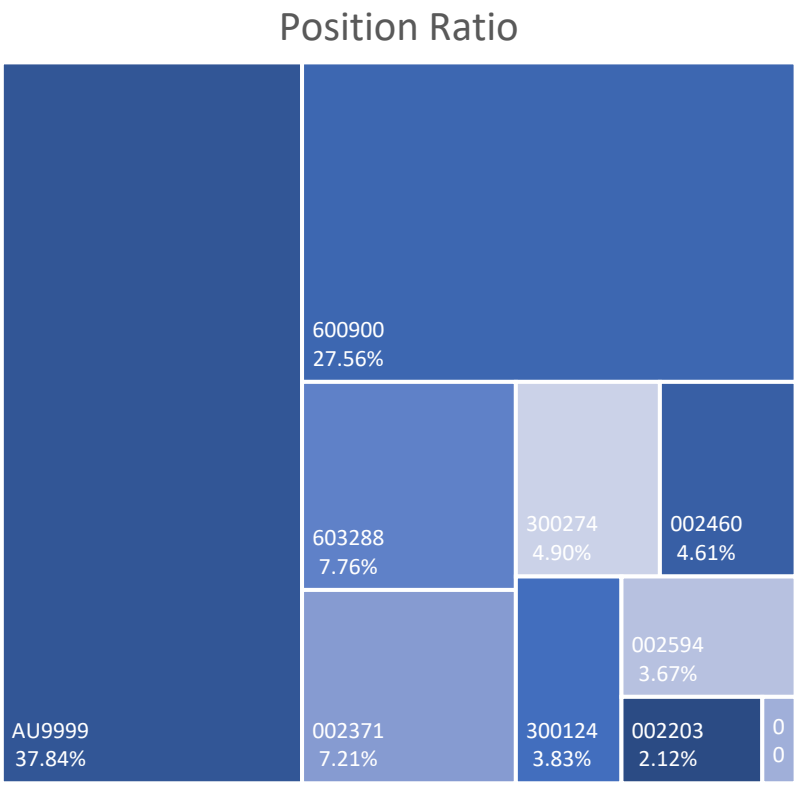
1. Portfolio Performance Evaluation

1.1 Overall Introduction

We have long position in 10 financial assets, 9 of them are equities and 1 is commodity (Gold).

Asset Name	Code
Sun Grow Power Co., Ltd	300274
Build Your Dreams	002594
Changchun High and New Technology Industries (Group) Inc	000661
NAURA Technology Group Co., Ltd	002371
Foshan Haitian Flavouring and Food Company Ltd	603288
Shenzhen Inovance Technology Co., Ltd	300124
China Yangtze Power Co., Ltd	600900
Ganfeng Lithium Group Co., Ltd	002460
Gold ETF	AU9999
Zhejiang Hailiang Co.,Ltd	002203



The asset we hold most is gold and the second-most is China Yangtze Power Co.,Ltd. Those two assets provide high and stable yield over time.

1.2 Efficient Frontier and Portfolio Ratio Analysis

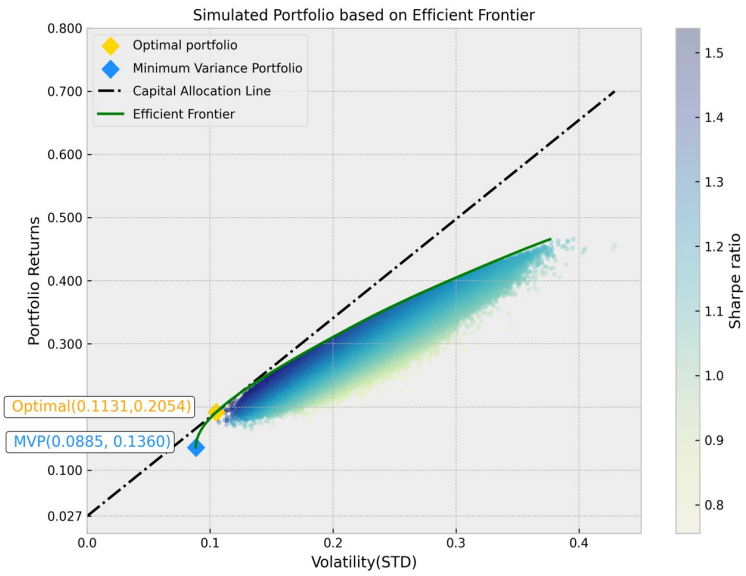


Exhibit: Efficient Frontier and CAL

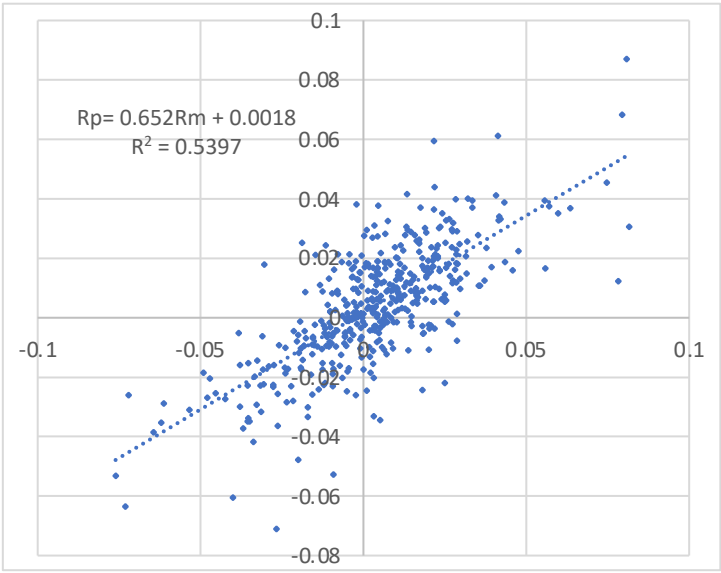
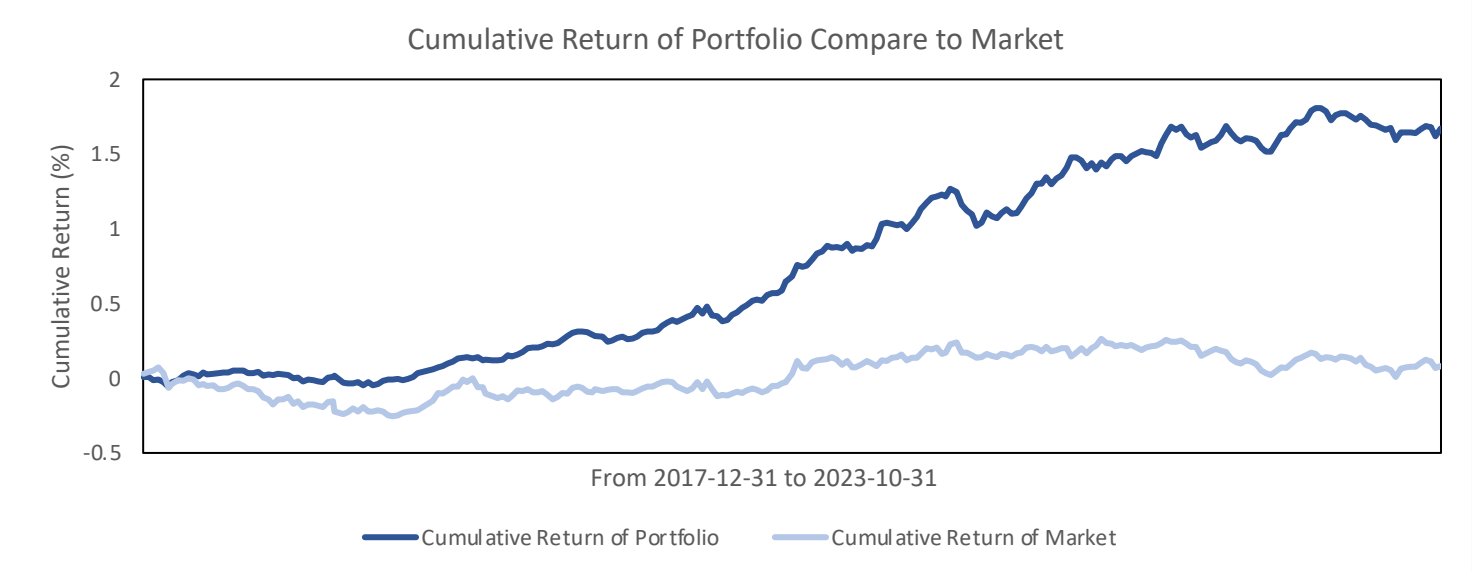


Exhibit: Single Index Model of Portfolio

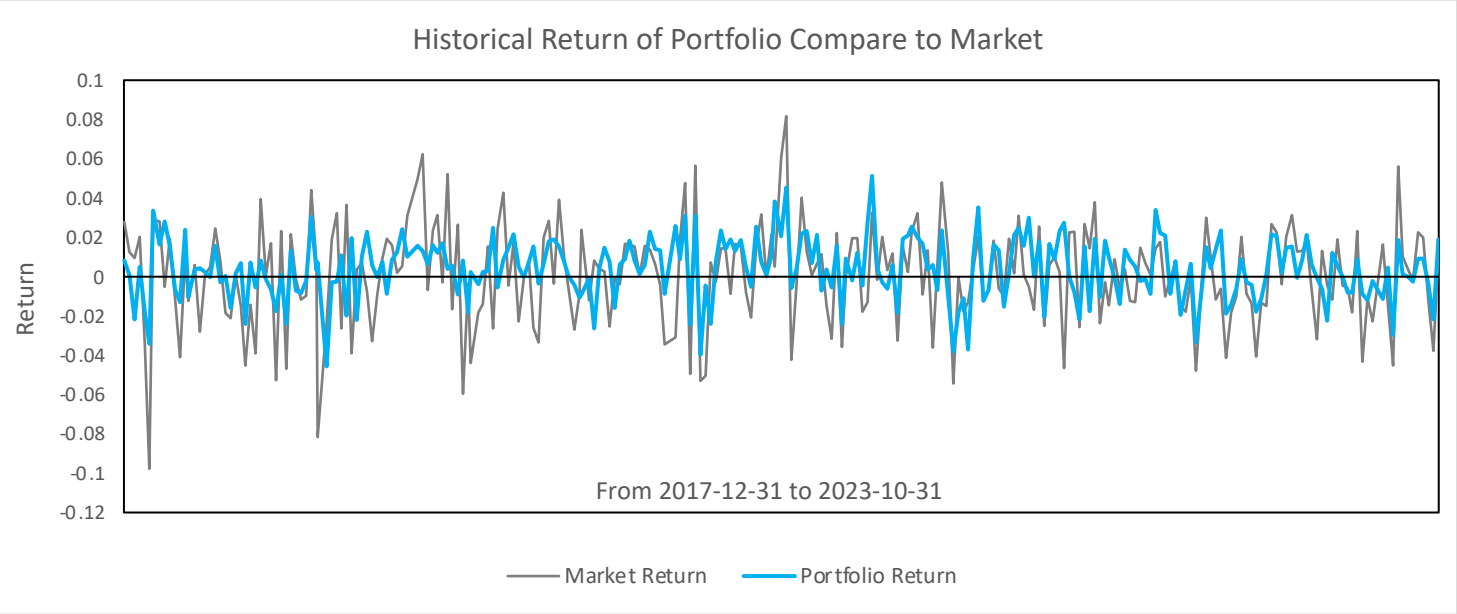
	Annual Return	Annual Risk	Sharpe Ratio	I-R(Single)	I-R(FF)	Alpha (Single)	Alpha(FF)
Portfolio	21.17%	11.72%	1.576	0.9529	1.6703	9.14%	21.92%
Market	3.24%	18.33%	-0.010	0	0	0	0

It is obvious to see that out portfolio has a smaller risk than market while providing higher yield. The Sharpe ratio of our portfolio is 1.576 while the Sharpe ratio of market is -0.01. The information ratio calculated from both single index model and Fama-French model are greater than 0, which means our portfolio generates positive return for unsystematic risk. Eventually, the annual alpha of our portfolio is 9.14% and 21.92% from two models respectively, showing that our portfolio is undervalued and having ability to earn abnormal return for our investors

1.3 Portfolio Performance compared to Market and Other Fund

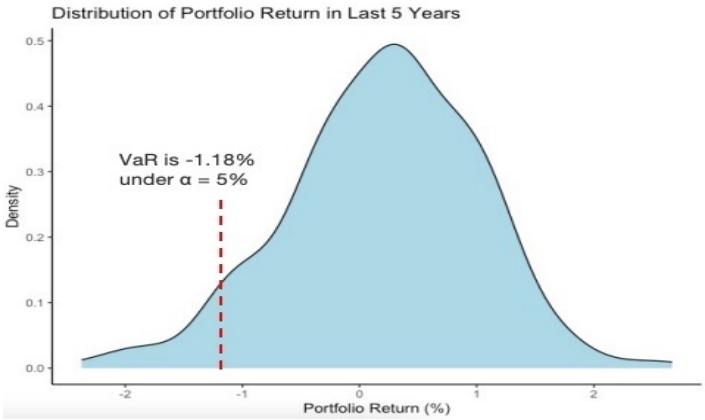


In the past 5 years, the cumulative return of our portfolio is higher than the cumulative return of market. The divergence between us and market is increasing over time. We strongly believe that our portfolio will have outstanding performance in the long run, offering stable and prominent yield to investors.



Since in the previous section, we already introduce the volatility of our portfolio, which is smaller than the market. We here want to introduce the Beta of our portfolio; Beta is an important indicator that measures the sensitivity of portfolio to market. From exhibit, it is easy to notice that our portfolio is not very sensitive to the market, which might be good news to those who are risk averse.

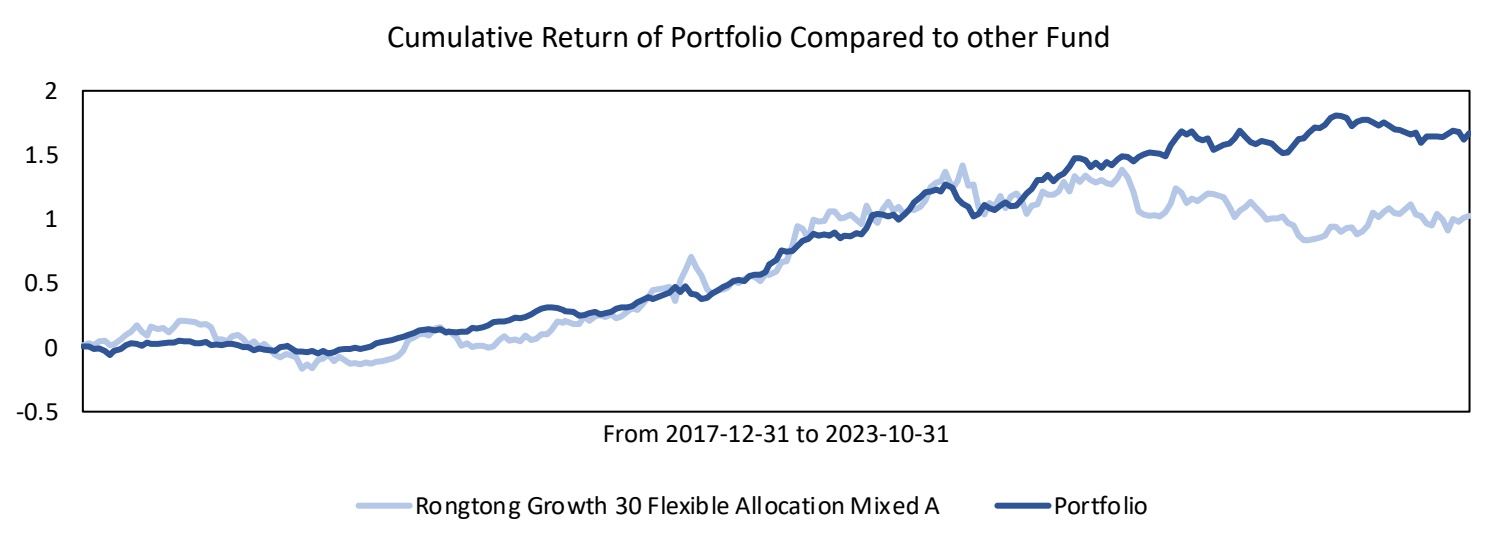
1.4 Value at Risk



Based on the historical performance of our portfolio in last 5 years, the Value at Risk at significance level of 5% is -1.18%(annualized). Which means there is a 5% chance that the investors may incur a loss of more than -1.18%.

This is a risk reminder to any investors who intend to have investment into this portfolio.

1.5 Portfolio Performance compare with other Funds



We select "Rongtong Growth30 Flexible Allocation Mixed A" as our comparison fund, whose major composition are Gold related stock (4.2%) and China Yangtze Power Co., Ltd (7.01%). In the past 5 years, the cumulative return of two funds are almost overlapped with a high momentum, showing that our selected securities have potential to grow. Since 2021, our portfolio still perform better than their fund with a cumulative return over 180% in 5 years. Therefore, we firmly believe our portfolio would be competitive related to other fund in the market.

Measurements Comparison

Performance	Portfolio	Rongtong Growth 30 Flexible Allocation Mixed A
Alpha	21.92%	13.69%
Beta	0.149	0.184
Sharpe Ratio	1.576	0.720
Treynor Ratio	1.239	0.785
Information Ratio	1.670	0.574
Annual Return	27.140%	17.182%
Annual STD	11.720%	23.853%

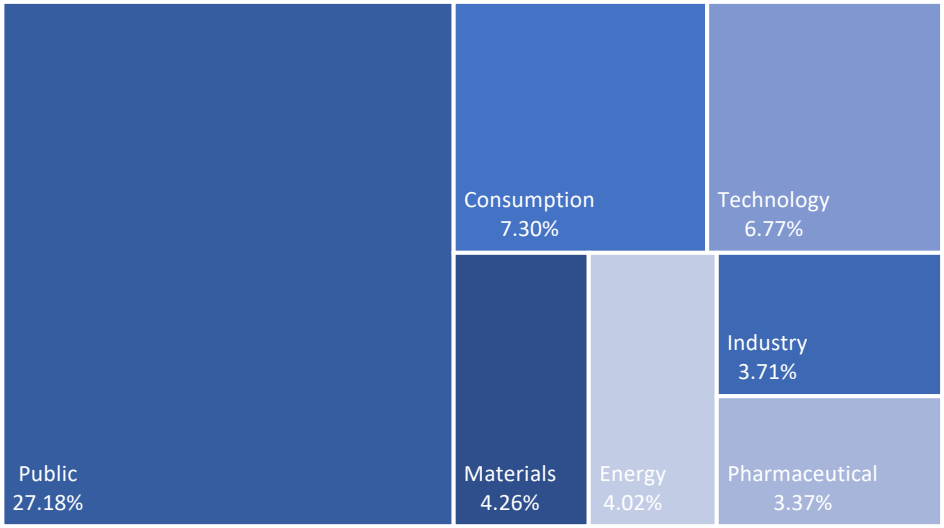
Compared to "Rongtong Growth30 Flexible Allocation Mixed A", our portfolio is more underpriced and less sensitive to market change.

With a higher Sharpe ratio, Treynor ratio and Information ratio, we generally believe that our portfolio will have a less fluctuation and much better performance compared to other similar funds.

2. Single Security Performance Evaluation

2.1 Industry Analysis

Sector Allocation



In our portfolio, expect **gold**, we choose 9 different stocks in 7 different industries. Others are from technology, consumption, industry and public industries. The corresponding sector allocation in our portfolio is shown.

Industry Correlation Matrix

	Energy	Pharmaceutical	Technology	Consumption	Industry	Public	Materials
Energy	1.0000						
Pharmaceutical	0.3895	1.0000					
Technology	0.3874	0.6641	1.0000				
Consumption	0.4900	0.7278	0.5502	1.0000			
Industry	0.6940	0.5909	0.5909	0.6153	1.0000		
Public	0.6936	0.4599	0.4909	0.4897	0.7499	1.0000	
Materials	0.7719	0.5802	0.5912	0.6309	0.7901	0.6610	1.0000

We use **Shanghai securities composite index** in industries to calculate the correlation between the different industries. This correlation table shows there are medium positive correlation industries in the investment portfolio. The performance of one industry may to some extent compensate for the losses of another industry that performs less well, which will diversification benefits.

Stock Correlation Matrix

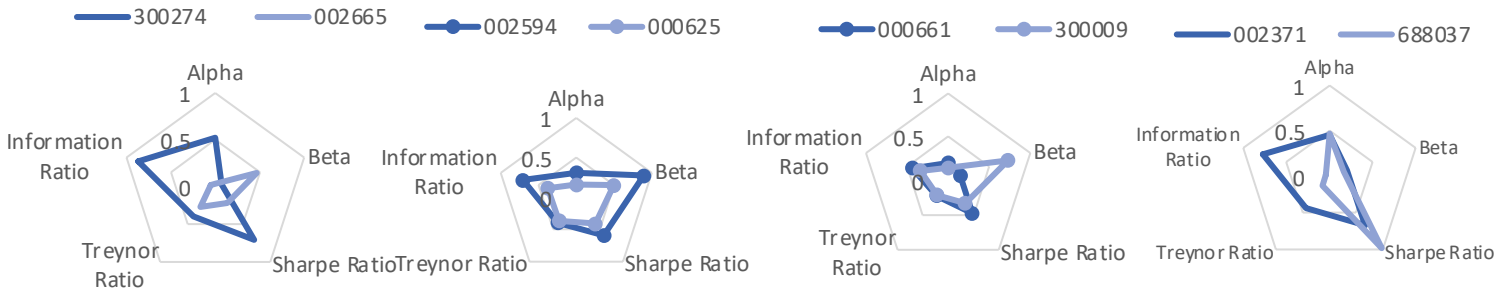
	300274	002594	000661	002371	603288	300124	600900	002460	002203
300274	1.0000								
002594	0.4787	1.0000							
000661	0.1712	0.1216	1.0000						
002371	0.3068	0.2631	0.2004	1.0000					
603288	0.1026	0.2312	0.3965	0.0931	1.0000				
300124	0.4641	0.4823	0.2276	0.2471	0.2983	1.0000			
600900	0.0027	0.0634	0.0415	-0.0475	0.1102	0.0194	1.0000		
002460	0.1750	0.0800	0.0714	0.0958	0.0170	0.1539	-0.0498	1.0000	
002203	0.1453	0.1528	0.0554	0.0645	0.0844	0.2017	0.0259	0.1455	1.0000

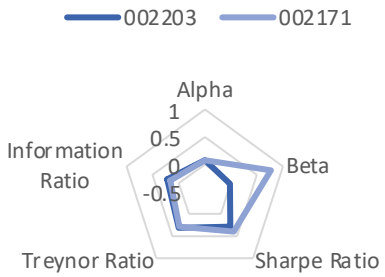
We also calculate the correlation between each stock. The correlation coefficient of these 9 stocks is basically less than 0.5, which indicates that they are weakly correlated and conducive to risk hedging.

2.2 Single Security Introduction

Stock Code	Introduction
300274	Focused on the solar energy industry, is one of the world's leading photovoltaic companies, committed to providing clean energy solutions.
002594	A comprehensive technology enterprise, covering electric vehicles, batteries, new energy and other fields.
000661	A leading biopharmaceutical enterprise in China, focusing on biopharmaceutical, medical devices.
002371	A high-tech enterprise specializing in optical communication and semiconductor fields, committed to providing advanced integrated circuit solutions.
603288	A leading enterprise in China's food industry, with traditional condiments such as soy sauce and vinegar as its leading products.
300124	Focusing on the field of industrial automation, providing servo control, industrial Internet and other solutions.
600900	Mainly engaged in hydropower, thermal power and other energy development and power generation business.
002460	Focusing on lithium materials and lithium battery industry.
002203	Focusing on the field of medicine and health products, with biopharmaceutical and medical devices as the leading business.

2.3 Radar Maps of comparing similar stocks





Stock Code	(Stock's EPS)* (Competitor's P/E)	Stock Price
300274	58.5488	54.2499
2594	184.6642	142.2761
661	604.40181	233.49
2371	340.0551	163.6279
603288	37.11692	62.7564
300124	47.7922	46.0001
600900	20.0321	18.9814
2460	178.6621	54.5568
2203	13.62547	9.754

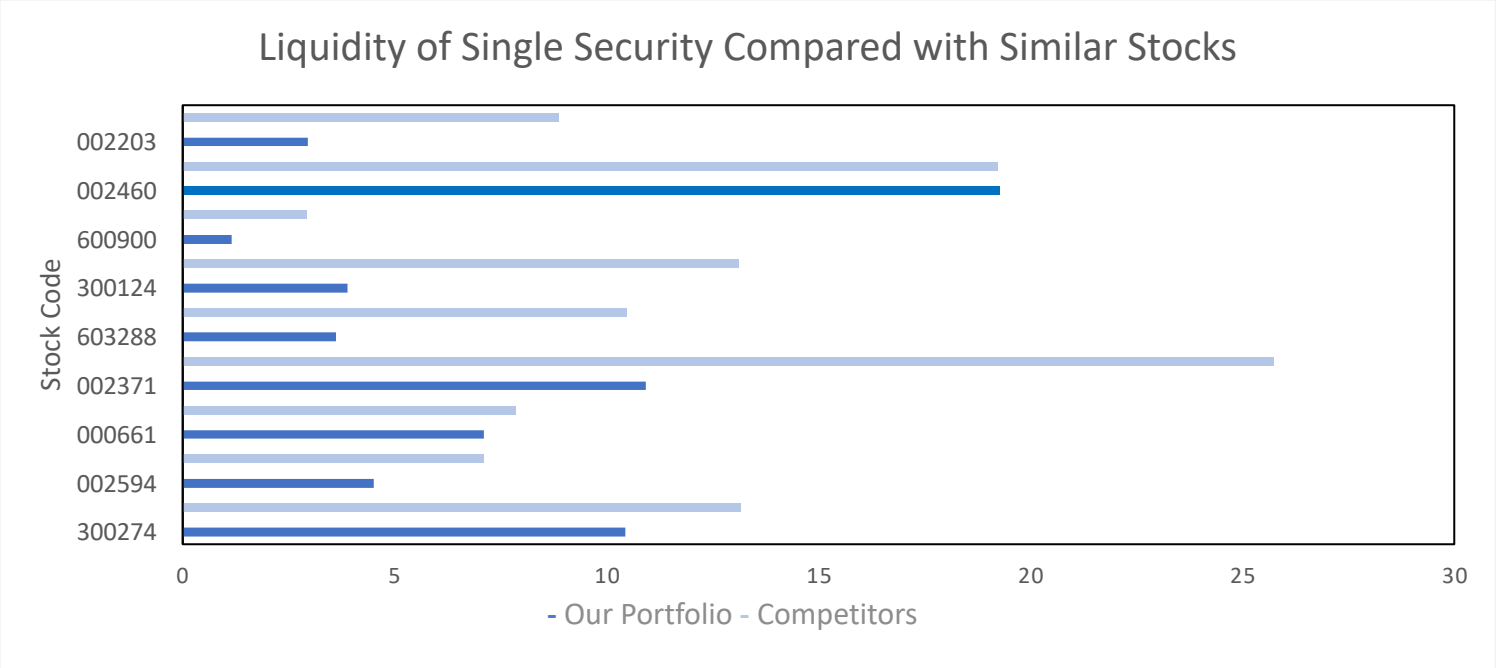
Exhibit: 5 Measurements of Single stocks
*Dark Blue are our stocks

(Stock's EPS) (Competitor's P/E) = Stock Intrinsic Value
Stock Price = Actual Price

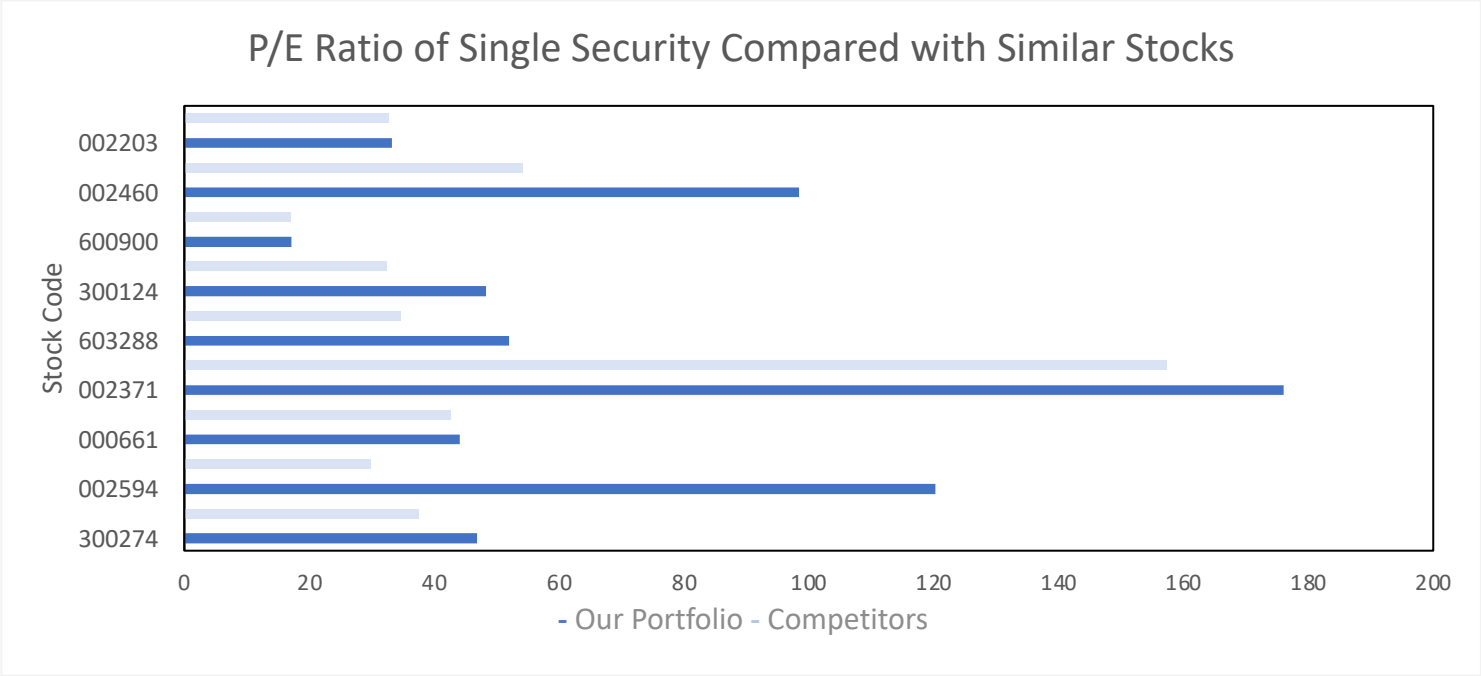
Each single securities in our portfolio generally has a higher alpha compared to similar competitors, showing the higher possibility of underprice, while the beta is relatively low compared to their competitors, which means that the sensitivity of our single security is relatively low to their competitors. Our selected securities' Sharpe, Treynor, Information ratios are all higher than competitors, displaying a better performance.

Applying P/E Method to compare the intrinsic value with the actual stock price, we may conclude that a large percentage of our selected stocks are underpriced.

2.4 Liquidity & P/E Plots of Comparing Similar Stocks



Based on the average of liquidity in the past five years, our competitors' liquidity generally perform better than our portfolio, displaying that our competitors are traded relatively more frequently in a short-term basis. Although our portfolio may be less liquid, it proves that most shareholders prefer holding our securities in a long-run. Therefore, we may conclude that our selected securities have a higher growth potential compared to similar stocks.



Based on the average P/E ratio compared to similar stocks in the past 5 years, we may stick to our conclusion that our selected securities all have highly growth potential, displaying the strong momentum of our portfolio.

3. Capital Allocation (Risk Preference)

3.1 Risk Lover

For risk lover people, we suggest they investing all money into our portfolio at optimal point with an annual return of 20.54%. Therefore, they can maximize their return and be well paid off from their investment.

3.2 Risk Averse Investor

For risk averse people who are afraid of risk during the investment, investing all the funds into our equity portfolio may not be suitable for them. In this way, we suggest this group of investors to have a clear mind about their required risk premium to variance ratio in advance and reallocate the portfolio for risk aversion investor.

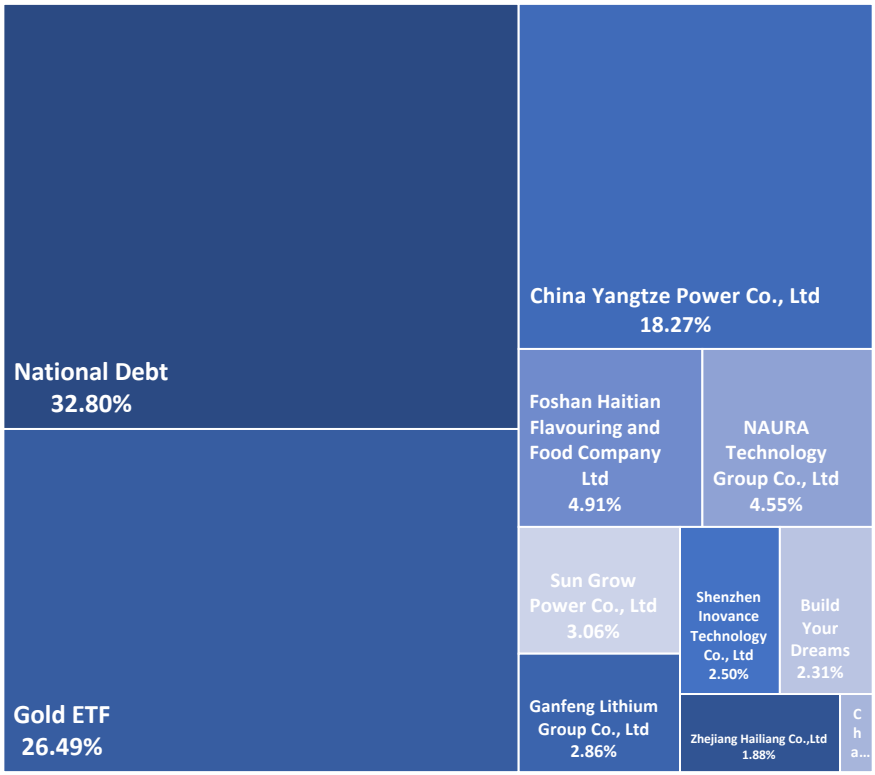
Mean of portfolio	0.2117
STD of portfolio	0.1172
Risk free rate	0.0270
Sharpe ratio	1.5756
ARPTVR	13.4393
RRPTVR	20.0000
Portfolio Proportion	0.6720
National Debt Proportion	0.3280

It is obvious to see that the available risk premium to variance ratio **(ARPTVR)** of our portfolio is 13.4393. If the investor required risk premium to variance ratio **(RRPTVR)** is higher than available risk premium to variance ratio, reallocation of the portfolio through add risk-free asset such as national debt may be necessary. For example, if the required risk premium to variance ratio is 20, we will suggest our investors to allocate 32.803% money into national debt while 67.197% into our equity portfolio.

Portfolio	
Asset	Weight
Sun Grow Power Co., Ltd	0.0306
Build Your Dreams	0.0231
Changchun High and New Technology Industries (Group) Inc	0.0039
NAURA Technology Group Co., Ltd	0.0455
Foshan Haitian Flavouring and Food Company Ltd	0.0491
Shenzhen Inovance Technology Co., Ltd	0.0250
China Yangtze Power Co., Ltd	0.1827
Ganfeng Lithium Group Co., Ltd	0.0286
999 Gold ETF	0.2649
Zhejiang Hailiang Co.,Ltd	0.0188
National Debt	0.3280
Total	1

Following is the example of completed portfolio to demonstrate the capital allocation and the composition of each assets in the final portfolio for risk averse investors.

CAPITAL ALLOCATION FOR RISK AVERSE INVESTOR



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